Vectors in $\mathbb{R}^3$

1. Vectors and their vector structure

Def. A vector $\vec{v}$ in space is characterized by three data:
- magnitude / length ($\|\vec{v}\|$)
- direction
- sense

Obs. Vectors are added by the triangle law or the parallelogram law.

Addition postulates:

1) $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
2) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3) $\exists\, \vec{0}$ s.t. $\vec{0} + \vec{v} = \vec{v}$
4) $\exists\, (-\vec{v})$ s.t. $\vec{v} + (-\vec{v}) = \vec{0}$

Denote: $\mathcal{V}_3$ the set of all vectors in space.

Obs. Then $(\mathcal{V}_3, +)$ is a commutative group.

Scalar multiplications:

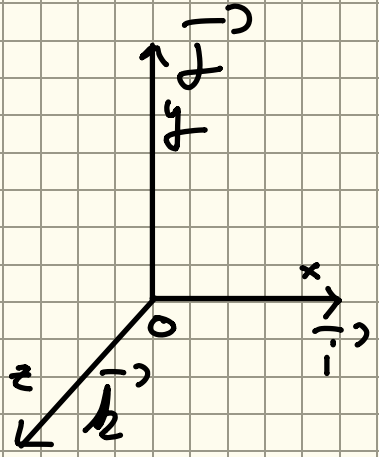
1) $1 \cdot \vec{a} = \vec{a}$
2) $\alpha \cdot (\beta \cdot \vec{a}) = (\alpha\beta)\vec{a}$
3) $(\alpha + \beta)\vec{a} = \alpha\vec{a} + \beta\vec{a}$
4) $\alpha(\vec{a} + \vec{b}) = \alpha\vec{a} + \alpha\vec{b}$

Obs. Numbers α, β are called scalars. $\beta\vec{v}$ is a

scalar multiple of $\vec{v}$.

Def. Fix mutually perpendicular axes Ox, Oy, Oz forming a right-handed frame, with unit vectors $\vec{i}$, $\vec{j}$, $\vec{k}$. Every vector $\vec{v}$ is written as:

$$\vec{v} = a\vec{i} + b\vec{j} + c\vec{k}, \quad a, b, c \in \mathbb{R}$$



Properties:

1) $\|a \cdot \vec{v}\| = |a| \cdot \|\vec{v}\|$
2) $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$

2. Inner product

Def. The scalar (inner) product is the number:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \cos\theta, \quad \theta = (\vec{a}, \vec{b})$$

Obs. With $\vec{a} \cdot \vec{b} = 0$ whenever $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$

Properties:

1) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
2) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
3) $(\alpha \cdot \vec{a}) \cdot \vec{b} = \alpha(\vec{a} \cdot \vec{b})$
4) $\vec{a} \cdot \vec{a} \geq 0 \iff \vec{a} = 0$

Obs. Since $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$:

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}, \qquad \vec{a} \cdot \vec{b} = 0 \iff a \perp b$$

$$\vec{i}^2 = \vec{j}^2 = \vec{k}^2 = 1$$
$$\vec{i} \cdot \vec{j} = \vec{i} \cdot \vec{k} = \vec{j} \cdot \vec{k} = 0$$

} The basis is orthonormal

Obs. Let $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$:

$$\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

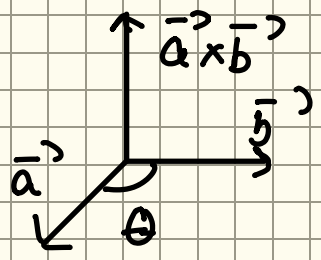
$$\cos\theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\cdot\sqrt{b_1^2 + b_2^2 + b_3^2}}$$

$$a \perp b \Leftrightarrow a_1b_1 + a_2b_2 + a_3b_3 = 0$$

3. Vector (cross) product

Def. The vector product $\vec{a}\times\vec{b}$ is the vector characterized by:

1) $\|\vec{a}\times\vec{b}\| = \|\vec{a}\|\cdot\|\vec{b}\|\cdot\sin\theta$
2) $(\vec{a}\times\vec{b}) \perp \vec{a}$ and $(\vec{a}\times\vec{b}) \perp \vec{b}$
3) $\{\vec{a}, \vec{b}, \vec{a}\times\vec{b}\}$ is oriented as $\{\vec{i}, \vec{j}, \vec{k}\}$ (right-hand rule)



Properties:

1) $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$
2) $(\alpha\cdot\vec{a})\times\vec{b} = \vec{a}\times(\alpha\cdot\vec{b}) = \alpha(\vec{a}\times\vec{b})$
3) $\vec{a}\times(\vec{b}+\vec{c}) = \vec{a}\times\vec{b} + \vec{a}\times\vec{c}$
4) $\vec{a}\times\vec{0} = \vec{0}$

5) $\vec{a} \times \vec{a} = \vec{0}$

6) $\vec{a} \times \vec{b} = \vec{0} \iff \vec{a} \parallel \vec{b}$

7) $\| \vec{a} \times \vec{b} \|$ = area of the parallelogram of $\vec{a}$ and $\vec{b}$.

Multiplication table for $\vec{i}, \vec{j}, \vec{k}$:

$\times$	$\vec{i}$	$\vec{j}$	$\vec{k}$
$\vec{i}$	0	$\vec{k}$	$-\vec{j}$
$\vec{j}$	$-\vec{k}$	0	$\vec{i}$
$\vec{k}$	$\vec{j}$	$-\vec{i}$	0

$$\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2)\vec{i} + (a_3 b_1 - a_1 b_3)\vec{j} + (a_1 b_2 - a_2 b_1)\vec{k}$$

Determinant formula:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

4. Triple product identity:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\cdot \vec{c}$$

Obs. This has no geometrical meaning. The product always lies in the plan spanned by $\vec{b}$ and $\vec{c}$ and in consistent with the right hand side.

5. Triple scalar product

Def. The triple scalar product of $\vec{a}, \vec{b}, \vec{c}$ is:

$$(\vec{a}, \vec{b}, \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Properties:

1) $(\vec{a}, \vec{b}, \vec{c}) = (\vec{c}, \vec{a}, \vec{b}) = (\vec{b}, \vec{c}, \vec{a})$

2) $(\vec{a}, \vec{b}, \vec{c}) = -(\vec{a}, \vec{c}, \vec{b})$

3) $(\alpha \cdot \vec{a}, \vec{b}, \vec{c}) = \alpha (\vec{a}, \vec{b}, \vec{c})$

4) $(\vec{u} + \vec{v}, \vec{b}, \vec{c}) = (\vec{u}, \vec{b}, \vec{c}) + (\vec{v}, \vec{b}, \vec{c})$

Obs. 1) $|(\vec{a}, \vec{b}, \vec{c})|$ is the volume of the parallelepiped spanned by $\vec{a}, \vec{b}, \vec{c}$.

2) $(\vec{a}, \vec{b}, \vec{c}) = 0 \iff \vec{a}, \vec{b}, \vec{c}$ coplanar

3) $(\vec{a}, \vec{b}, \vec{c})$ has positive orientation if $(\vec{a}, \vec{b}, \vec{c}) > 0$

4) $(\vec{a}, \vec{b}, \vec{c})$ has negative orientation if $(\vec{a}, \vec{b}, \vec{c}) < 0$.